



ABB's SWIFTI at work (Courtesy: ABB)

ing an item from point A to point B, but it makes its own decision on the best way to make that delivery. Lastly, cobots are armed with a degree of sensing and data collection that were not available to industrial robots. Cobots can collect data from a warehouse floor, for example, and have that data fed into a larger automation strategy. Industrial robots were stand-alone devices, without a larger view of the automation strategy.”

Open Robotics is the primary maintainer of the Robot Operating System (ROS). ROS gained popularity after its utilisation by many of the robots at the Defense Advanced Research Projects Agency (DARPA) Robotics Challenge, which began in 2012. NASA used it on its Robonaut 2 humanoid robot and Astrobe systems, which are deployed on the International Space Station (ISS), and in the VIPER, a mobile robot that NASA aims to send to the South Pole of the Moon in 2024.

Safe to work with humans.

ISO/TS 15066:2016 specifies safety requirements for collaborative industrial robot systems and the work environment, and supplements the requirements and guidance on collaborative industrial robot operation given in ISO 10218-1 and ISO 10218-2.

Suraj K.K., Senior Sales Engineer, Systemantics India Pvt Ltd, an Indian company that has designed and developed collaborative robotic arms indigenously from the ground up, explains that one or more of the following requirements must be met

to ensure that humans are not exposed to unacceptable risks.

One, the robot stops when humans enter the collaborative space. Two, its motion is controlled by a human hand. Three, the robot maintains a specific distance and speed. Or four, its inherent design and controls limit impact at contact events. “Our products comply with the

second and fourth point, partially complying with the third,” he says.

Different applications require different levels of collaboration—and different levels of collaboration require different levels of safety.

Guruprasad of ABB explains further, “There are three types of collaboration: coexistence, intermittent collaboration, and cooperative or close collaboration. Coexistence means safe and shared use of the same workspace between the cobots and humans without protective fences, for example, palletising. Synchronised collaboration means planned but more intermittent interaction between the cobots and humans, such as in machine tending. Cooperation is the highest level of collaboration where workspace and tasks are shared continuously, for example, on small parts of an assembly line. Cobots enable human interventions multiple times during the robot operational cycle, for example, where a part of the assembly of a component is done by the cobot and another part by a human. The safety functions of ABB cobots are according to ISO10218-1 and ISO/TS 15066. In order to account for the unpredictability of human behaviour and movements, ABB’s collaborative robots are enabled with safety functions such as power and force limiting, speed and separation monitoring, hand guiding and safety-rated stop.”

“Cobots go through various safety and security measures before they are deemed safe to be used in factories.

Each machine passes through various tests and safety standards. Cobots come with robotic vision such as sensors and laser cameras. Robotics arms are integrated with laser cameras and sensors to sense their surroundings and work accordingly. Technology has advanced in the last few decades, and artificial intelligence (AI) has developed to a great extent that if a human comes in contact with a machine working area, the machine stops automatically and saves humans from any potential injury,” says Sougandh.

Balancing safety and functionality. Suraj remarks that a user’s perspective of an ideal cobot will depend on where the emphasis lies—safety, functionality, or both. The safety features explained above have to be balanced with functional requirements like ease-of-use (which also translates to flexibility), support and service, weight of the robot, and accuracy and repeatability (depending on the area of application).

Compact and lightweight. In addition to the smaller footprint, lower weight, ease-of-use, and safety make cobots more versatile than regular industrial robots. They can be deployed even in places with space constraints, comfortably placed near workers, to assist them without eating into their workspace. “A collaborative robotic arm can be mounted on a pedestal to service a cell, a floor gantry to service a line, or a vehicle to service an area,” says Suraj. Cobots can be easily moved around and set up for multiple applications within the same plant. Generally, they require minimal or no infrastructural modifications to the work floor.

Adaptable to multiple tasks. Sougandh of UR says, “Traditional robots need engineers to operate them, and their functions are predetermined. Cobots are adaptable and capable of carrying out many activities. Each machine is built to carry out a range of functions and is simple to program. Even operators with limited or no robotics or programming background